



Session 02: Unit Conversions & Dimensional Analysis

Topic: Moving between units without getting lost. The chain method.

Time: 150 minutes

Big Fat Notebook: Chapters 1–2

Opening Attack

Try this:

A gas pump at an American station dispenses **12.5 gallons** of gasoline. The density of gasoline is **0.720 g/mL**.

How many **milliliters** is 12.5 gallons?

And given that density — what is the mass in **kilograms**?

Stuck on where to start? Good. By the end of this session, you'll chain conversions like dominoes.

Worked Pattern

Example 1 — Single-Step Metric Conversion

Problem: Convert 3500 mL to liters.

Step 1: Find the conversion factor.

$$1 \text{ L} = 1000 \text{ mL}$$

Step 2: Write as a fraction so the unwanted unit cancels.

$$3500 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = ?$$

Step 3: Multiply. mL cancels, L remains.

$$3500 \times 1 \text{ L} \div 1000 = 3.5 \text{ L}$$

Answer: 3.5 L (2 sig figs, from 3500 — ambiguous, but 3.5×10^3 would be clearer).

Example 2 — Single-Step: Which Way Does the Fraction Go?

Problem: Convert 0.450 kg to grams.

The question: should I multiply by $\frac{1000 \text{ g}}{1 \text{ kg}}$ or $\frac{1 \text{ kg}}{1000 \text{ g}}$?

Rule: Put the unit you **WANT** on top. Put the unit you **HAVE** on bottom.

I HAVE kg. I WANT g.

→ $\frac{1000 \text{ g}}{1 \text{ kg}}$ ← g on top, kg on bottom. Correct.

$$0.450 \text{ kg} \times \frac{1000 \text{ g}}{1 \text{ kg}} = 450 \text{ g}$$

$$\cancel{\text{kg}} \times \frac{\text{g}}{\cancel{\text{kg}}} = \text{g} \checkmark$$

Answer: 450 g (3 sig figs).

Example 3 — Multi-Step Chain Conversion

Problem: Convert 12.5 gallons to milliliters.

Given: 1 gal = 3.785 L, 1 L = 1000 mL

Step 1: Plan the chain: gallons → liters → milliliters.

Step 2: Write each conversion so unwanted units cancel.

$$12.5 \text{ gal} \times \frac{3.785 \text{ L}}{1 \text{ gal}} \times \frac{1000 \text{ mL}}{1 \text{ L}}$$

Step 3: Cancel and multiply.

$$\cancel{\text{gal}} \times \frac{\cancel{\text{L}}}{\cancel{\text{gal}}} \times \frac{\text{mL}}{\cancel{\text{L}}} = \text{mL}$$

$$12.5 \times 3.785 \times 1000 = 47,312.5 \text{ mL}$$

Step 4: Sig figs. 12.5 has 3 sig figs. 3.785 has 4. 1000 mL/L is exact.

$$\rightarrow 47,300 \text{ mL} \rightarrow 4.73 \times 10^4 \text{ mL (3 sig figs)}$$

Answer: $4.73 \times 10^4 \text{ mL}$ (3 sig figs).

Example 4 — Multi-Step with Mass from Density

Problem: Using the 12.5 gallons from Example 3 (which we found is $4.73 \times 10^4 \text{ mL}$), find the mass in kilograms. Density = 0.720 g/mL.

Step 1: Use density as a conversion factor: $\frac{0.720 \text{ g}}{1 \text{ mL}}$.

Step 2: Chain: mL → g → kg.

$$4.73 \times 10^4 \text{ mL} \times \frac{0.720 \text{ g}}{1 \text{ mL}} \times \frac{1 \text{ kg}}{1000 \text{ g}}$$

Step 3: Cancel and multiply.

$$\cancel{\text{mL}} \times \frac{\cancel{\text{g}}}{\cancel{\text{mL}}} \times \frac{\text{kg}}{\cancel{\text{g}}} = \text{kg}$$

$$4.73 \times 10^4 \times 0.720 \div 1000 = 34.056 \text{ kg}$$

Step 4: Sig figs. 0.720 has 3 sig figs → 34.1 kg.

Answer: 34.1 kg (3 sig figs).

Example 5 — Derived Unit Conversion: km/h → m/s

Problem: A car travels at 90.0 km/h. Convert to m/s.

Step 1: Plan: km/h → m/h → m/s. Two conversions needed: km → m, and h → s.

Step 2: Write the chain.

$$\frac{90.0 \text{ km}}{1 \text{ h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}}$$

Step 3: Cancel and multiply.

$$\frac{\cancel{\text{km}}}{\cancel{\text{h}}} \times \frac{\text{m}}{\cancel{\text{km}}} \times \frac{\cancel{\text{h}}}{\text{s}} = \frac{\text{m}}{\text{s}}$$

$$90.0 \times 1000 \div 3600 = 25.0 \text{ m/s}$$

Answer: 25.0 m/s (3 sig figs, from 90.0).

Note: The conversion 1 h = 3600 s is exact. Sig figs come from 90.0.

Example 6 — Density as a Conversion Factor (Both Directions)

Problem: Gold has density 19.3 g/cm³.

(a) What is the mass of 5.00 cm³ of gold?

(b) What is the volume of 100.0 g of gold?

(a) Mass from volume:

$$5.00 \text{ cm}^3 \times \frac{19.3 \text{ g}}{1 \text{ cm}^3} = 96.5 \text{ g}$$

$$\cancel{\text{cm}^3} \times \frac{\text{g}}{\cancel{\text{cm}^3}} = \text{g} \checkmark \rightarrow 96.5 \text{ g (3 sig figs)}$$

(b) Volume from mass — flip the density:

$$100.0 \text{ g} \times \frac{1 \text{ cm}^3}{19.3 \text{ g}} = 5.181... \text{ cm}^3$$

$$\cancel{\text{g}} \times \frac{\text{cm}^3}{\cancel{\text{g}}} = \text{cm}^3 \checkmark \rightarrow 5.18 \text{ cm}^3 \text{ (3 sig figs, limited by 19.3)}$$

Answers: (a) 96.5 g (b) 5.18 cm³

Example 7 — Squared/Cubed Unit Conversion

Problem: A floor is 15.0 m². Convert to cm².

The trap: 1 m = 100 cm, so many students write:

$$\cancel{15.0 \text{ m}^2 \times \frac{100 \text{ cm}}{1 \text{ m}}} = \cancel{1500 \text{ cm}^2} \leftarrow \text{WRONG!}$$

Why? Because m² means m × m. You need to square the conversion:

$$\left(\frac{100 \text{ cm}}{1 \text{ m}} \right)^2 = \frac{100^2 \text{ cm}^2}{1^2 \text{ m}^2} = \frac{10,000 \text{ cm}^2}{1 \text{ m}^2}$$

Correct:

$$15.0 \text{ m}^2 \times \frac{10,000 \text{ cm}^2}{1 \text{ m}^2} = 150,000 \text{ cm}^2 = 1.50 \times 10^5 \text{ cm}^2$$

Answer: 1.50 × 10⁵ cm² (3 sig figs).

The conversion squared is NOT 100 — it's 10,000. This is the most common unit conversion error in chemistry.

Example 8 — Volume Units: cm³ ↔ mL ↔ L

Problem: A container holds 250 cm³. How many liters is this?

Key fact (memorize this): $1 \text{ cm}^3 = 1 \text{ mL}$. Always. Exactly.

Step 1: $\text{cm}^3 \rightarrow \text{mL}$ (same number).

$$250 \text{ cm}^3 = 250 \text{ mL}$$

Step 2: $\text{mL} \rightarrow \text{L}$.

$$250 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 0.250 \text{ L}$$

Answer: 0.250 L (3 sig figs).

THE TRAP — Forgetting to Square/Cube the Conversion Factor

The wrong way:

Convert 3.0 m^3 to cm^3 .

$$\text{Wrong work: } 3.0 \text{ m}^3 \times \frac{100 \text{ cm}}{1 \text{ m}} = 300 \text{ cm}^3$$

This is off by a factor of **1,000,000**.

Why is this wrong? $\text{m}^3 = \text{m} \times \text{m} \times \text{m}$. If $1 \text{ m} = 100 \text{ cm}$, then $1 \text{ m}^3 = (100 \text{ cm})^3 = 1,000,000 \text{ cm}^3$.

The right way:

$$3.0 \text{ m}^3 \times \left(\frac{100 \text{ cm}}{1 \text{ m}} \right)^3 = 3.0 \text{ m}^3 \times \frac{1,000,000 \text{ cm}^3}{1 \text{ m}^3} = 3.0 \times 10^6 \text{ cm}^3$$

Spot the difference: Whenever the unit has an exponent (m^2 , m^3), the conversion factor gets the SAME exponent. $100^2 = 10,000$. $100^3 = 1,000,000$.

What We Just Did — The Dimensional Analysis Chain

Step 1: Write the starting quantity with its unit. Identify the target unit.

"I have ____ in (unit A). I want (unit B)."

Step 2: Build a chain of conversion fractions.

Each fraction puts the unit you **WANT** on top and the unit you're **CANCELING** on bottom.

Multiply across. Cancel units at every step.

$$\cancel{\text{unit A}} \times \frac{\text{unit B}}{\cancel{\text{unit A}}} \times \frac{\text{unit C}}{\cancel{\text{unit B}}} = \text{unit C}$$

Step 3: When the only unit left is the target unit, multiply all top numbers, divide by all bottom numbers.

Round to correct sig figs. If the unit has an exponent, the conversion factor gets that same exponent.

Copy this. This is the procedure behind every unit conversion you will ever do in chemistry.

Pro Tips — Calculator Shortcuts & Mental Checks

Tip 1 — The "want on top, have on bottom" rule. Never guess which way the fraction goes. If you have kg and want g, write $\frac{\text{g}}{\text{kg}}$. The unit you're canceling always goes on bottom.

Tip 2 — $1 \text{ cm}^3 = 1 \text{ mL}$. Always. Exactly. This is the single most useful volume conversion in

chemistry. A 500 cm³ container holds exactly 500 mL. Never convert — just swap the label.

Tip 3 — Density flips both ways. Need mass from volume? $m = \rho V$. Need volume from mass? $V = m/\rho$. The density fraction just flips. Don't memorize two formulas — just use dimensional analysis and let the units tell you which way it goes.

Tip 4 — km/h → m/s shortcut. Divide by 3.6. $90.0 \div 3.6 = 25.0$ m/s. Reverse: m/s → km/h, multiply by 3.6. This saves 10 seconds on every speed conversion.

Tip 5 — Squared/cubed sanity check. If you're converting cm² to m² and get a number bigger than what you started with, you forgot to square. 1 m² = 10,000 cm² — converting TO m² should give a MUCH smaller number. 15,000 cm² = 1.5 m², not 150 m².

Basic Drills (B1–B10)

Goal: Automate the chain. Units must cancel cleanly every time.

B1. Convert 5.00 km to meters.

B2. Convert 450 mg to grams.

B3. Convert 2.50 hours to seconds. (1 h = 60 min, 1 min = 60 s — both exact)

B4. Convert 65.0 mi/h to km/h. (1 mi = 1.609 km)

B5. A bottle holds 2.00 L. Convert to mL, then to cm³.

B6. Convert 0.850 g/mL to kg/L.

B7. A gold ring has volume 0.450 cm³. Gold's density is 19.3 g/cm³. Find mass in grams, then in kilograms.

B8. A swimming pool holds 50.0 m³ of water. Convert to liters. (1 m³ = 1000 L)

B9. (Two-part) Part 1: Convert 100.0 yards to meters (1 yd = 0.9144 m). Part 2: Convert that same distance to centimeters.

B10. (Speed round — under 2 min) Convert 3.00×10^8 m/s (speed of light) to km/h. Use

scientific notation.

Advanced Drills (A1–A10)

Goal: Multi-step, traps, exam time pressure.

A1. (Multi-step integration with Session 01) A car travels 425.0 km on 32.5 L of fuel. (a) Convert km to miles (1 km = 0.6214 mi). (b) Convert L to gallons (1 gal = 3.785 L). (c) Calculate fuel economy in miles per gallon with correct sig figs.

A2. (Multi-step integration) A recipe calls for 2.50 cups of flour. 1 cup = 236.6 mL. Flour density = 0.593 g/mL. Find the mass of flour in pounds (1 lb = 453.6 g).

A3. (Trap-laden — squared conversion) A square plot of land is 250. m on each side. (a) Calculate area in m². (b) Convert to km². (c) A student does: $250 \times 250 = 62,500 \text{ m}^2$, then $62,500 \times (1/1000) = 62.5 \text{ km}^2$. What mistake did they make? Show the correct answer.

A4. (Reverse engineering) A student converts 45.0 m/s to km/h by writing: $45.0 \times (1 \text{ km}/1000 \text{ m}) \times (3600 \text{ s}/1 \text{ h}) = 162 \text{ km/h}$. Is this correct? If yes, explain why the 3600 went on top. If no, find the mistake.

A5. (Constructive) Write an exam problem that requires exactly 3 conversion steps and ends with a mass in kilograms. Use at least one English→metric conversion. Solve it with full dimensional analysis.

A6. (Constructive) Write a problem where a student who forgets to cube the conversion factor gets an answer that is off by a factor of exactly 1,000,000. Solve both the wrong way and the right way.

A7. (Table-based) A shipping manifest lists the following items:

Item	Volume (cm ³)	Density (g/cm ³)
Steel bolt	8.50	7.85
Aluminum washer	2.40	2.70
Copper wire	1.15	8.96

Calculate the total mass in kg. ($1\text{ kg} = 1000\text{ g}$)

A8. (Timed — under 3 min) A cylindrical tank has radius 1.50 m and height 4.00 m. Volume = $\pi r^2 h$. It is filled with a liquid of density 0.880 g/mL. Find the mass of the liquid in metric tons. (1 metric ton = 1000 kg. $1\text{ m}^3 = 10^6\text{ cm}^3 = 10^6\text{ mL}$.)

A9. (Hardest variation) The "jer" is a made-up unit: $1\text{ jer} = 3.50\text{ meters}$. The "flerp" is another: $1\text{ flerp} = 12.0\text{ seconds}$. Convert 45.0 jer/flerp to km/h. This tests whether you understand the structure of derived units.

A10. (Exam boss) A NASA contractor mixes up units. They calculate a satellite's mass as 150. kg and report it to a partner who assumes the number is in pounds (lb). The partner uses 150. lb in calculations. ($1\text{ kg} = 2.205\text{ lb}$). (a) What mass in kg did the partner actually use? (b) If the satellite needs at least 140. kg of fuel and the partner loads fuel based on their incorrect mass at a ratio of 0.200 kg fuel per 1 kg satellite mass, how much fuel (in kg) do they actually load? (c) Is it enough? Show all work in a clear chain.

Now Read

You can now convert any unit to any other unit using dimensional analysis — at exam speed.

Read **Chapters 1–2** of *Everything You Need to Ace Chemistry in One Big Fat Notebook*.

The concepts — why unit cancellation works, how density bridges mass and volume, why squared conversions are the #1 trap — will lock into place. You've solved 20 problems harder than anything the book asks.

Solutions: <solutions/02-solutions.md>



Key Terms — Quick Reference

Term	What it means
dimensional analysis	the chain method — multiply by conversion fractions until units cancel
conversion factor	a fraction equal to 1 (e.g., $\frac{1000 \text{ mL}}{1 \text{ L}}$)
derived unit	unit made from base units: km/h, g/mL, g/cm ³
metric prefixes	k (10 ³), c (10 ⁻²), m (10 ⁻³), μ (10 ⁻⁶)
density (ρ)	$\rho = m/V$; bridge between mass and volume
1 cm³ = 1 mL	exactly — always, no exceptions
squared/cubed conversion	if the unit has an exponent, the conversion factor gets the same exponent: $(100)^2 = 10,000$
want/have rule	unit you WANT on top, unit you HAVE on bottom
exact numbers	conversions like 1 min = 60 s have infinite sig figs — they never limit precision
density flip	$m = \rho V$ or $V = m/\rho$ — the fraction flips based on what you're solving for